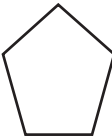


Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Both DNA and RNA are polynucleotides.

(a) The table shows how the components of polynucleotides can be represented.

Component	Representation
sugar	
phosphate group	
cytosine or thymine	
adenine or guanine	
covalent bond	
hydrogen bond	

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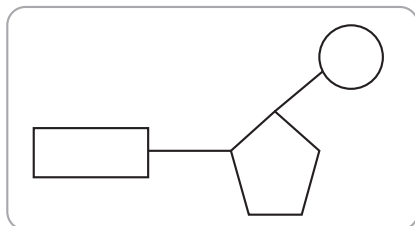
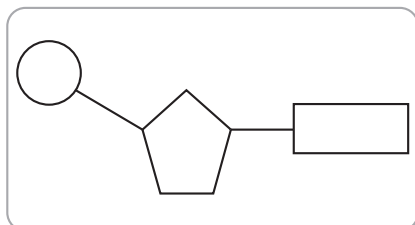
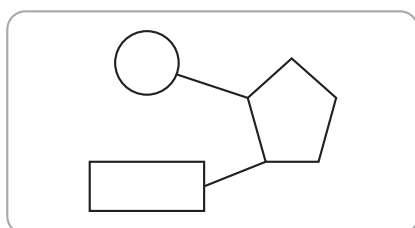
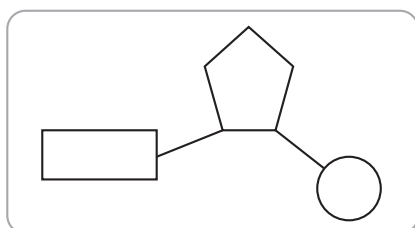
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(i) Which diagram shows a mononucleotide?

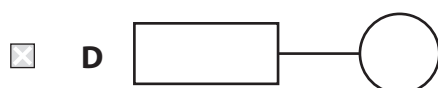
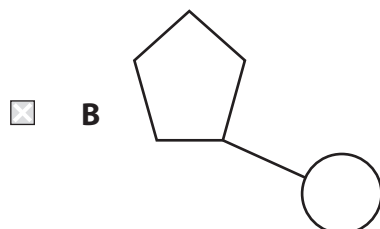
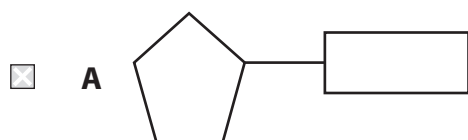
(1)

**A****B****C****D**

P 6 6 1 5 3 A 0 3 3 2

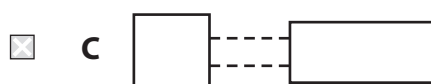
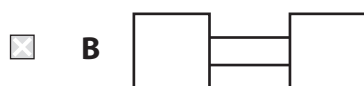
(ii) Which diagram shows two components joined by a phosphodiester bond?

(1)



(iii) Which diagram shows complementary base pairing?

(1)



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(b) The diagram shows a sequence of bases in a DNA template (antisense) strand.

A	T	T	G	G	C	G	A	G	A	A	C
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- (i) The table shows some statements about the new complementary DNA strand and the mRNA strand synthesised using this sequence of bases.

For each statement, put **one** cross ☒ in the appropriate box, in each row, to show the correct statement about these strands.

(3)

Statement	both the new complementary DNA strand and the mRNA strand	only the new complementary DNA strand	only the mRNA strand	neither strand
The number of guanines will be the same as in the template strand	☒	☒	☒	☒
The number of thymines will be the same as the number of adenines in the template strand	☒	☒	☒	☒
There will be no adenine present	☒	☒	☒	☒

- (ii) Name the process that synthesises a mRNA strand using the DNA template strand.

(1)

(Total for Question 1 = 7 marks)

